extra information on the needs and vulnerability of different sections of the population, fueling a case for conservation where protection is required. As reliable information on age ranges and breeding rates is often not available for small, threatened populations, ecologists sometimes use data from other populations of the same species—or a different but similar species—to conduct a PVA. However, the results are variable, even in populations of the same species in the same area. In a 2015 study of three colonies of California sea lion in the Gulf of California, "surrogate" data from one colony was used to make forecasts on the other two; they proved valid for one colony, but not the other.

"We can't just do conservation: we have to meet all these other needs from the land, too. What recovery is enough?"

Georgina Mace

Making a difference

Methods are still being refined, but PVA has now become a cornerstone of conservation biology. PVAs have been applied to populations as varied as island foxes in California, sea otters in Alaska, Fender's blue butterflies and Northern Spotted Owls in Oregon, and bottlenose dolphins off the coasts of Argentina and Australia. With the development of increasingly efficient computer programs incorporating ever more variables, PVA will undoubtedly be used even more effectively in the future. It is impossible to predict every extinction, but PVA provides tools for identifying endangered populations and determining the management actions likely to be most effective in improving population viability, and preserving a species at risk.